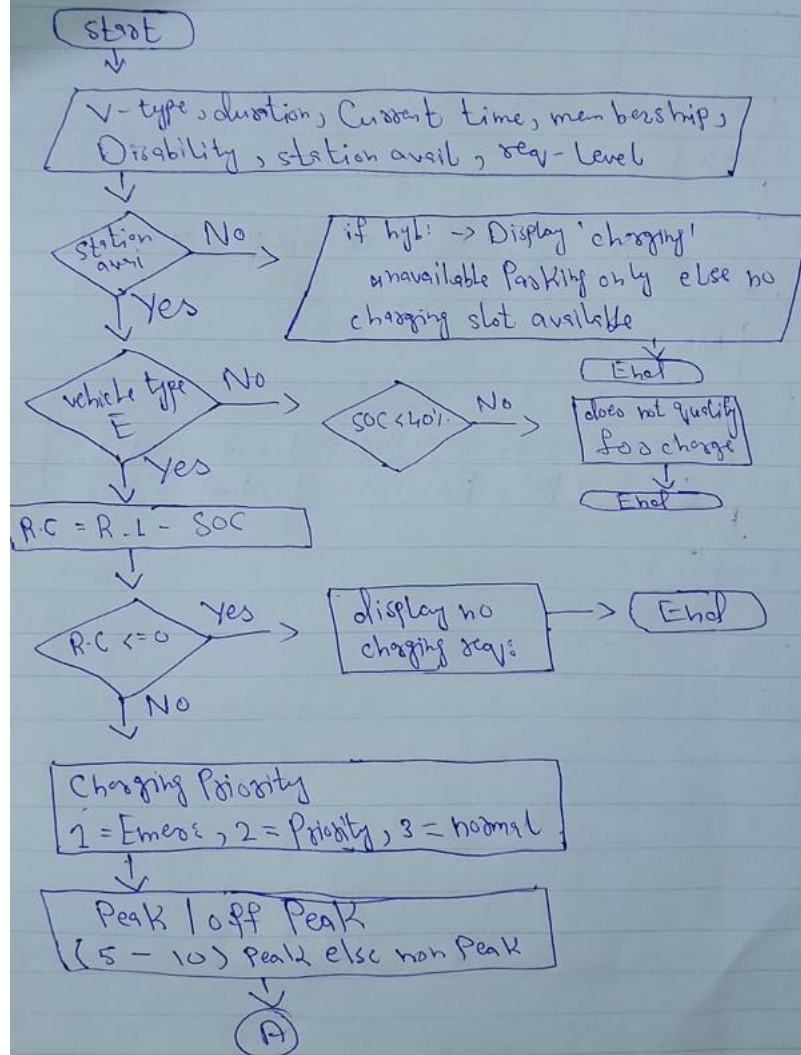
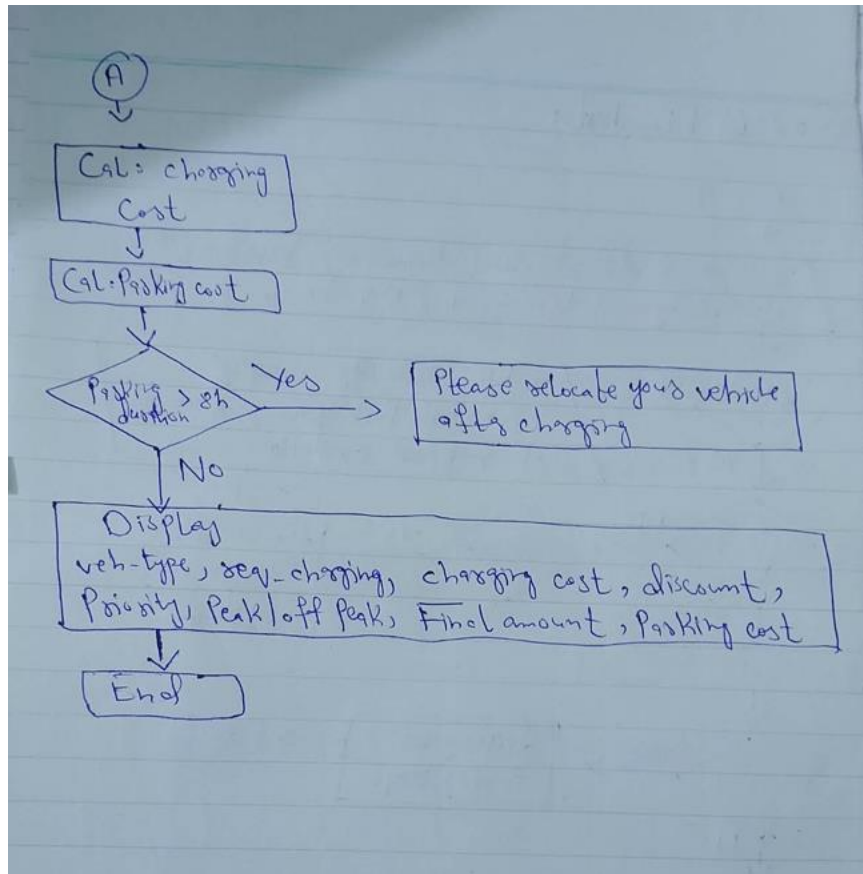


Q#6(b) Flowchart:





Pseudocode:

```
1. start
2. input
3. Read V-type, SOC, Req. Level, Duration, arrival time
4. // check station availability
5. If station-avail = 'N' then
6.   If V-type = 'H' then
7.     Print "charging unavailable, parking only"
8.   else
9.     Print no charging slot available
10.  End if
11.  Go to end
12. End if
13. // check vehicle eligibility
14. if V-type = 'E' then
15.   Allow charging = true
16. Else if V-type = 'H' and SOC < 40 then
17.   Allow charging = true
18. Else
19.   Print "vehicle does not qualify for charging"
20.   Go to end
21. End if
22. // check Required charging
23. req: charging = Req. Level - SOC
24. if req: charging <= 0 then
25.   Print "no charging require"
26.   Go to end
```

```

27. End if
28. // Determine Priority
29. if current time < 17 or current time ≥ 22 then
30.     Peak = "off-Peak"
31. Else
32.     Peak = "Peak"
33. End if
34. // calculate charging cost
35. if Priority = "Emergency" then
36.     charging cost = rate * require charging
37. else
38.     charging cost = rate * require charging
39. End if
40. Calculate Parking Cost
41. if disability = 1 then
42.     Personal cost = 0
43. Else if duration ≤ 2 then
44.     P.C = 200
45. Else if duration ≤ 5 then
46.     P.C = 400
47. Else
48.     Parking-cost = 700
49. End if
50. if membership = 'Y' and disability < 1 then
51.     P.C = P.C * 0.80
52. End if
53. Final Amount and Message

```

Date: _____

```

54. Final amount = charging-cost + Parking cost
55. if duration > 8 then
56.   Print "Long stay warning"
57. Else
58.   Print "standard Parking duration"
59. End if
60. 11 Display result:
61.   Print vehicle type, SOC, Req-Level,
62.   Priority, Peak/off Peak, charging cost, Parking
63.   cost, discount, final-amount
64. End
  
```

I/O-Chart

Input	Process	Output
• Vehicle type (I/h)	• Check station availability	• vehicle type
• Battery charge (SOC)	• Verify vehicle	• Current Battery %
• Req. charging level	• Cal: R.C	• Req. charging %
• Expected Parking duration	• Assign Prio:	• charging Priority
• Current time	• Peak/off Peak	• Peak/off Peak
• P. membership	• Cal: C.C	• C.C
• Disabled-person	• Cal: P.C	• P.C
• Priority	• Apply discount	• Final amount
• charging station available	• Check long-stay warning	

PAC - Chart 8

Date: _____

Step	Description	Input	Process	Output
Read input	Get vehicle and customer detail	Vehicle type, Soc, Reg, Level, duration, time, membership, P, station avail	Vehicle and store	Validated data
Check station	Availability of charging station	Station avail, V-type	Decision	message or continue
Check eligibility	Vehicle Qualify for charging	Reg-type, SOC	Decision	allow reject
Cal: Req charge	need for charge	Req-level, SOC	Cal:	R.C
Assign Prio:	charging Prio:	Soc, Req-level, dir,	Decision	Priority
Peak/off Peak	Time based	Current time	Decision	Peak/off Peak
Cal: C.C	Charge amount	Req charge, Peak/off Peak membership	Cal:	C.C
Cal: Parking Cost	P.A	Duration, membership, disability	Cal:	P.C
Final amount	T.C	C.C, P.C discount	Cal:	F.A
Display result	show all details	All variables	Output	message